

hours of mobile and fixed-line internet shutdowns cost the country almost \$678 million.

The ICRIER study also noted that the Internet is the “primary driver of entrepreneurship”, and rightfully so. In today’s digitally-accelerated world, the internet enables everything, from a pan shop’s sizeable source of income (Paytm, GPay, Bhim) to the workings of unicorn startups! We’re targeting to become a \$5 trillion economy in the coming years, however, the rampant internet shutdowns send all the wrong signals to potential international investors and consumers.

Additionally, for the massive number of Indians who have put their trust in the vision of Digital India, internet blackouts threaten their socio-economic existence, or at least, stall it. Closer to home, due to our overdependence on the internet, we are used to Google-ing our way through almost every event in our life. We do not plan out trips in advance, do not research the routes for small journeys, and some of us don’t even carry physical cash on us anymore since the advent of UPI. An internet shutdown disrupts, disturbs and disorganises our lives to an enormous extent. Overall, policymakers need to create more stringent internet-related laws to protect the socio-economic interests of the citizens as well as the country.

WORKAROUNDS

In case of an internet shutdown, which could be a partial shutdown (particular apps such as Facebook, Twitter, etc) or complete shutdown, you can use:

- 1) Tools such as Psiphon, an open source web proxy, and browsers such as Tor, which prevent location tracing
- 2) A VPN which will make you bypass the restrictions set by the government.
- 3) Apps such as Orbot let you browse the internet anonymously
- 4) Using Tails, an operating system that allows you to bypass censorship and uses cryptographic tools to encrypt files
- 5) Lantern allows people in uncensored areas to share their internet
- 6) FireChat, an off-the-grid messaging app that lets users chat using Bluetooth. 

decent-ralise

Will DWeb (decentralised web) transform the internet as we know it today?

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The internet taketh as much as it giveth. This has become increasingly evident as the misuse of data, invasion of privacy, and government intervention, censoring and spying have become massive concerns for digitally-connected individuals across the world. In fact, we even had a cover story back in June 2018 which detailed how you could ‘Delete Yourself’ from the internet due to the rising privacy and data security concerns. Amidst the insecurities and agitation surrounding these issues, the idea of a new decentralised version of cyberspace, also known as the DWeb, is beginning to emerge. Certain proponents of this decentralised version of the web are also calling this Web 3.0.

WHAT LED TO DWEB?

Let’s briefly run through the two generations of the web, so far. The inception of the internet saw the first generation which lasted from 1989 to around 2005. It was mostly a passive medium with minimal interaction between users. A read-only version of the web, if you will, where the users were merely recipients of the disseminated information.



DWeb or decentralised web is being called Web3.0 by some proponents

Then came along Web 2.0, which ushered in social networks, wikis, blogs, online marketplaces and more which let web users create and share content, communicate with one other and share information through centralised services provided by tech giants such as Google, Microsoft, Facebook and Amazon. DWeb attempts to decentralise this process and relieve users from being reliant on these platforms. Users

will assume control over their data as they connect and exchange information directly with others in the network on a decentralised platform.

WAIT, THIS SOUNDS FAMILIAR...

Does all of this sound eerily similar to a little something known as blockchain? That’s because it is. The very first applications of blockchain (decentralised ledgers) technology were in the digital currency transactions area, including cryptocurrencies such as bitcoin and ethereum. The very same technology could now be applied to a larger scale operation – the web. Additionally, there are some cryptocurrencies which are also being deployed to create the DWeb such as Filecoin by Protocol Labs.

WHAT STIRRED THE INCEPTION OF DWEB

The need for this revolutionised web is evident. With Web 2.0, all user data across the world is accumulated by a few large organisations which make the risk of data hacking increase manifold. The centralised nature of our data also